

PROFESSIONAL POWER AMPLIFIER

TTA SERIES



TTA300 / TTA500

350–450W × 2 @ 8Ω
930–1460W Bridge

TTA600 / TTA800

650–850W × 2 @ 8Ω
1860–2500W Bridge

TTA1000 / TTA1300

1000–1300W × 2 @ 8Ω
3200–3900W Bridge

TTA450 / TTA460

450–650W × 4 @ 8Ω
4-Channel Models



DOCUMENT INFORMATION

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Models Covered	TTA300 TTA500 TTA600 TTA800 TTA1000 TTA1300 TTA450 TTA460
Manufacturer	Louis Martin Audio www.louismartinaudio.it

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1. Safety Instructions

- Read all instructions carefully before operating.
- Do not use this product near water or in damp environments.
- Clean only with a dry cloth. Never use liquid cleaners.
- Do not block ventilation openings. Maintain at least 1U (44mm) clearance above and below in racks.
- Do not install near heat sources (radiators, stoves, other amplifiers).
- Protect the power cord from pinching, especially at plugs and the unit entry point.
- Unplug during lightning storms or extended periods of non-use.
- All servicing must be carried out by qualified service personnel only.
- Ensure the mains supply voltage matches the amplifier's rated voltage (220–240V AC, 50/60Hz).
- Disconnect power before making or changing speaker connections.

NOTE: This unit contains no user-serviceable parts inside. Opening the chassis voids the warranty and may expose lethal voltages.

2. Product Overview

The TTA Series is a range of professional multi-channel Class TD power amplifiers engineered by Louis Martin Audio for demanding live sound, installation, and touring applications. Combining high power output with the efficiency of Class D and the sonic purity of Class B, intelligent thermal management, and comprehensive protection circuitry, the TTA Series delivers exceptional audio fidelity with outstanding reliability.

Key Features:

- Class TD amplifier topology — combines Class D efficiency with Class B sonic purity
- Tracks the audio signal at all frequencies, providing required rail voltage with additional headroom
- Automatic temperature-controlled variable-speed fan cooling system
- Multi-mode operation: Stereo, Parallel (4-ch models), and Bridge Mono
- Full protection: short-circuit, DC offset, over-temperature, over-current, clip limiting
- Balanced XLR inputs with 20k Ω impedance for noise-free long cable runs
- Heavy-duty gold-plated binding post speaker outputs per channel
- Signal, Clip and Protect LED indicators per channel
- 19-inch rack-mountable chassis (2U, 88mm height)
- Soft-start circuit with gradual volume increase to prevent speaker transients
- Fits both Touring Sound and Installation market requirements

3. Package Contents

- 1 × TTA Series Power Amplifier (specified model)
- 1 × IEC AC Power Cord
- 4 × Rack Mounting Screws (M6, with plastic washers)
- 1 × User Manual (this document)
- 1 × Warranty Registration Card

NOTE: Inspect the packaging for shipping damage before use. Contact your dealer immediately if any items are missing or damaged.

4. Front Panel Controls



Figure 1a — TTA Series Two-Channel Front Panel (TTA300–TTA1300)



Figure 1b — TTA Series Four-Channel Front Panel (TTA450 / TTA460)

Ventilation Slots	Perforated grille providing airflow for the internal heatsink and fan. Never obstruct.
Channel A / B Gain Knob	Continuously variable rotary attenuator (MIN to MAX). Controls individual channel output level. Fully clockwise = maximum gain.
Signal LED (Green)	Illuminates when audio signal is present above –40 dBu.
Clip LED (Orange)	Confirms active signal path. Illuminates when the output approaches clipping (approx. –3 dBFS).
Protect LED (Red)	Reduce source level or amp gain. Illuminates during a protection event: over-temperature, DC fault, short-circuit, or over-current. Unit mutes automatically.
Power Switch	Rocker switch (ON / OFF). Always power amplifiers on last in a signal chain and off first to prevent transient pops.

5. Rear Panel Connections



Figure 2a — TTA Series Two-Channel Rear Panel (TTA300–TTA1300)



Figure 2b — TTA Series Four-Channel Rear Panel (TTA450 / TTA460)

Cooling Fins / Fan Exhaust	Always ensure unobstructed airflow. Minimum 100mm clearance behind unit when rack-mounted.
AC Mains IEC Socket	Connect supplied power cord. The unit accepts 220–240V AC, 50/60Hz.
XLR Input CH A / CH B	Balanced XLR-3F connectors. Pin 1: GND, Pin 2: Hot (+), Pin 3: Cold (–). Accept signal levels from 0.775V to 1.44V (+4 dBu).
Output Binding Posts CH A/B	Heavy-duty gold-plated binding posts: Red (+) and Black (–) per channel. Speakon adapter compatible.
LINK / THRU (XLR)	Buffered signal loop-through output for chaining to additional amplifiers without signal degradation.
Bridge Output Terminals	Dedicated bridge connection between CH A (+) and CH B (–) for mono bridge operation. 2-channel models only. Selects amplifier operating mode:
Bridge / Mode Switch	STEREO or BRIDGE MONO. See Section 10 for full details.
GAIN Switch	Sets input sensitivity: 0.775V or 1.44V. Match to your source equipment for optimal noise performance.
Chassis Ground Screw	Earth bonding point. Connect to rack chassis ground for hum elimination in large installations.

6. Technical Specifications

Two-Channel Models (TTA300 – TTA1300)

Specification	TTA300	TTA500	TTA600	TTA800	TTA1000	TTA1300
Power @8Ω/ch	350W×2	450W×2	650W×2	850W×2	1000W×2	1300W×2
Power @4Ω/ch	465W×2	750W×2	930W×2	1250W×2	1600W×2	1950W×2
Bridge @8Ω	930W	1460W	1860W	2500W	3200W	3900W
Avg Current	3A/230V	4A/230V	4.5A/230V	5A/230V	5.5A/230V	6A/230V
Power Cons.	690W	920W	1035W	1150W	1265W	1380W
Net Weight	15 kg	17 kg	18 kg	20 kg	24 kg	26 kg

Four-Channel Models (TTA450 & TTA460)

Specification	TTA450	TTA460
Power @8Ω/ch	450W × 4	650W × 4
Power @4Ω/ch	750W × 4	930W × 4
Bridge @8Ω	—	—
Avg Current	6A @ 230V AC	7A @ 230V AC
Power Cons.	1380W	1610W
Net Weight	23 kg	25 kg

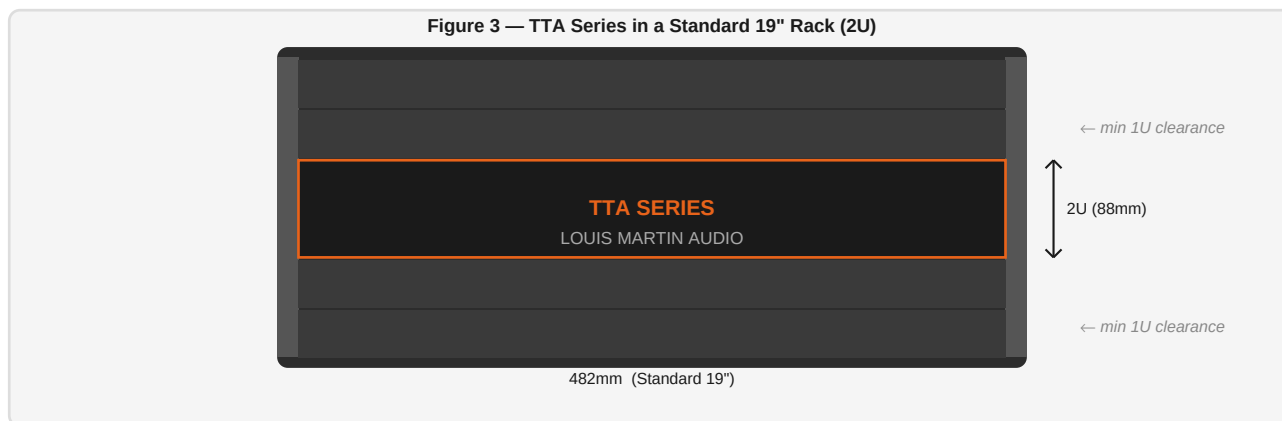
Common Electrical Parameters (All Models)

Parameter	Specification
Frequency Response	20Hz – 20kHz (±0.5 dB)
THD+N	<0.05% @ 8Ω, 1kHz
S/N Ratio (2-ch)	>98 dB (A-weighted)
S/N Ratio (4-ch)	>95 dB (A-weighted)
Damping Factor	>500 @ 1kHz/8Ω
Crosstalk (2-ch)	0.775V / 1.44V
Crosstalk (4-ch)	>75 dB @ 1kHz
Input Sensitivity	0.775V / 1.44V Default Gain: 32 dB
Input Impedance	20kΩ Balanced
Protection	Clipping, Thermal, Short Circuit, DC Offset, Over-Current
Dimensions (W×D×H)	483 × 442 × 88 mm / 483 × 462 × 88 mm

NOTE: All power ratings are continuous RMS at rated load impedance with <1% THD+N, both channels driven simultaneously at 1kHz unless otherwise stated.

7. Installation & Rack Mounting

Rack Mounting



The TTA Series amplifiers are designed for standard 19-inch equipment racks and occupy 2U (88mm) of rack space. Follow these steps for safe installation:

1. Ensure the rack is stable and level before installation.
2. Use all four supplied M6 rack screws with plastic washers to avoid damaging the rack ears.
3. Maintain at least 1U (44mm) of empty space above or below the amplifier for airflow.
4. The fan draws cool air from the front and expels hot air from the rear — ensure rear clearance of at least 100mm.
5. In dense rack configurations, consider a 1U ventilation panel above the amplifier.
6. Always route speaker cables away from signal cables to avoid induced noise and interference.
7. Connect the chassis ground screw to the rack frame if hum is encountered.

Standalone Operation

When not rack-mounted, place the amplifier on a stable, flat, ventilated surface. The rubber feet provide adequate clearance for bottom ventilation. Do not stack units directly on top of each other without rack mounting.

8. Wiring & Connection Guide

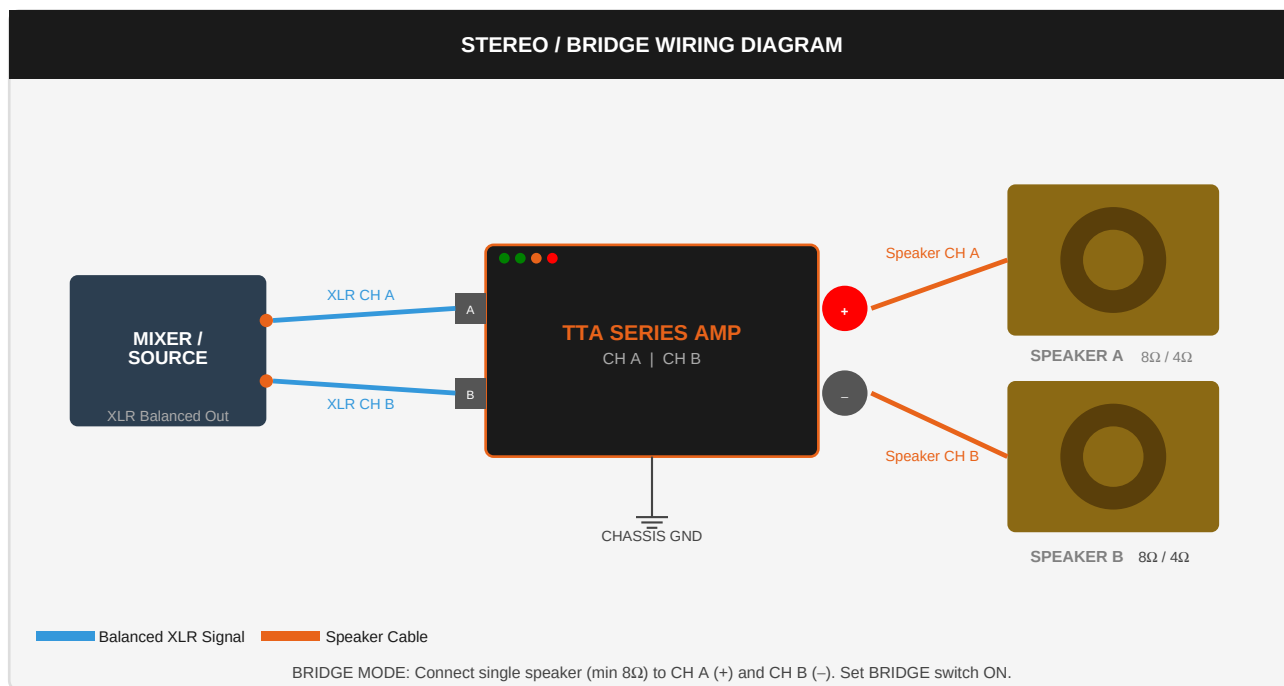
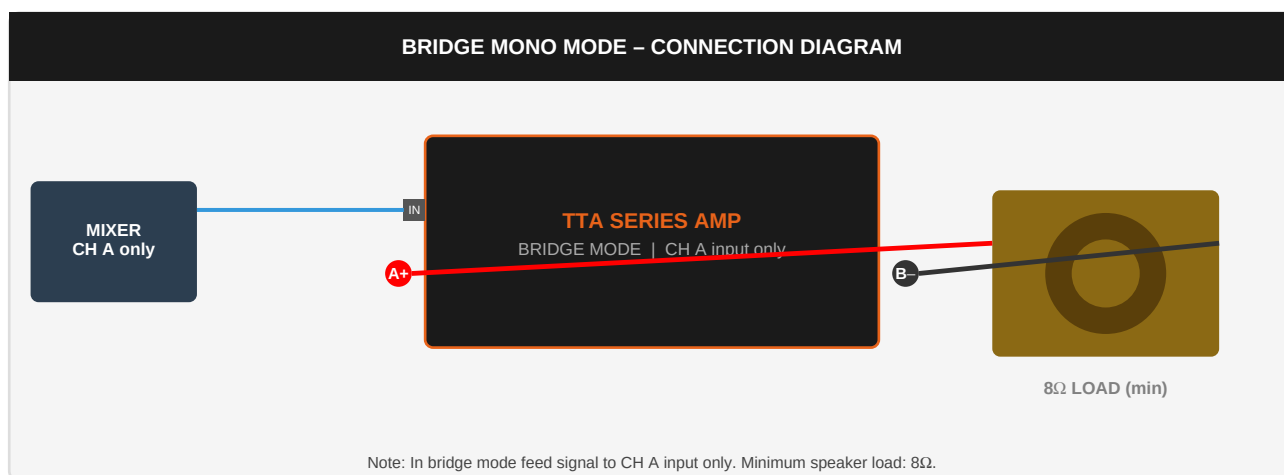


Figure 4 — Complete Stereo / Bridge Wiring Diagram

Basic Stereo Connection (Recommended for most installations)

1. Power OFF both the mixer/source and the amplifier before connecting.
2. Connect the mixer's balanced XLR output (CH L) to the amplifier's XLR Input CH A.
3. Connect the mixer's balanced XLR output (CH R) to the amplifier's XLR Input CH B.
4. Connect a speaker cable from Amplifier CH A (+/-) binding posts to the left speaker.
5. Connect a speaker cable from Amplifier CH B (+/-) binding posts to the right speaker.
6. Set the Bridge/Mode switch to STEREO.
7. Set gain knobs to minimum (fully counter-clockwise). Power on the amplifier last, then slowly raise gain.

Bridge Mono Connection



1. Power OFF all equipment before connecting.
2. Connect the mono source signal to CH A XLR input only.

3. Set the rear panel switch to BRIDGE.
4. Connect the speaker between the CH A (+) Red and CH B (–) Black binding posts. Minimum impedance: 8Ω.
5. Do NOT connect a speaker to the CH B output alone in bridge mode.
6. Power on the amplifier and raise the CH A gain knob to desired level.

■ **WARNING: Never connect a speaker load less than 8Ω in bridge mode. Operating below minimum impedance will trigger protection and may cause permanent damage to the output stage.**

Parallel Mode Connection (TTA450 / TTA460 four-channel models only)

In Parallel mode, all output channels carry the same signal sourced from CH A input. Useful for driving multiple speaker cabinets in distributed audio installations. Set the rear panel switch to PARALLEL. Connect source only to CH A XLR input. Minimum impedance: 4Ω per channel.

9. XLR & Speaker Output Pinout Diagrams

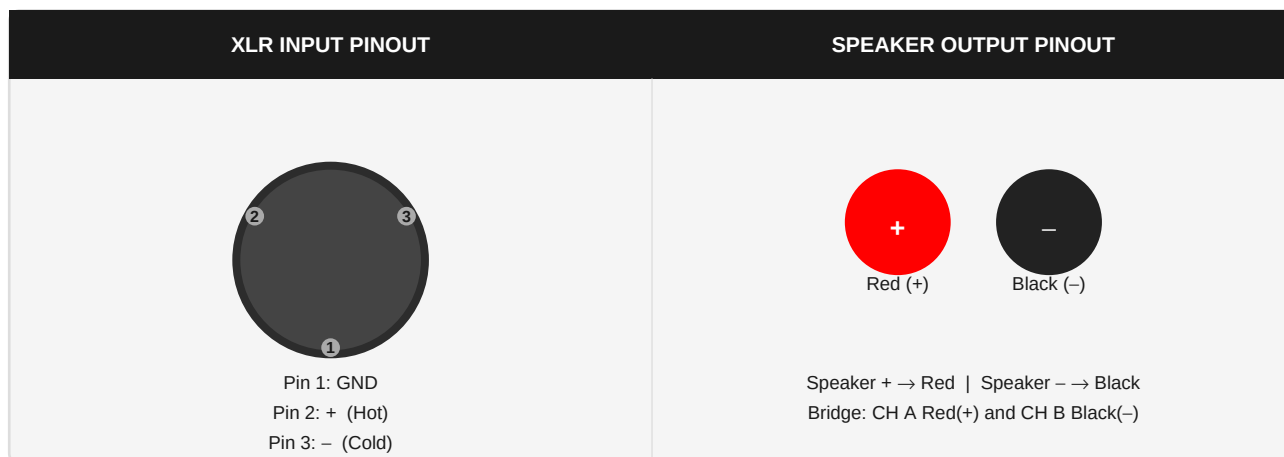


Figure 5 — XLR Input (left) and Speaker Output Binding Posts (right) Pinout Diagrams

XLR Input Wiring Notes

- Pin 1 (GND/Shield): Connect to cable shield. Lift at source end if ground hum occurs.
- Pin 2 (+): Hot / positive / non-inverting signal.
- Pin 3 (–): Cold / negative / inverting signal.
- For unbalanced (RCA/TS) sources, use an active DI box or balanced adapter. Do not leave Pin 3 floating.
- Maximum input level before clipping: +20 dBu (at 32 dB gain setting).

Speaker Output Wiring Notes

- Red (+) / Black (–): Standard binding post connection for all single-channel loads.
- Use minimum 2.5mm² conductor cross-section for speaker cables longer than 10 metres.
- In Bridge mode: connect speaker between CH A Red (+) and CH B Black (–) — minimum 8Ω.
- Never connect or disconnect speaker cables under signal — always mute or attenuate first.
- Ensure all binding post connections are tight and free from stray wire strands that could cause shorts.

10. Operating Modes

Stereo Mode

Normal operation. Each channel (A and B) receives an independent input signal and drives an independent speaker load. Both channel gain knobs operate independently. Minimum speaker impedance: 4Ω per channel.

Parallel Mode (TTA450 / TTA460 only)

CH A input is internally routed to all output channels simultaneously. Use when an identical signal is required across multiple speaker cabinets in installed sound or distributed audio. Set the rear panel switch to PARALLEL. Connect source only to CH A XLR input. Minimum impedance: 4Ω per channel (8Ω minimum if wiring speakers in parallel from one output).

Bridge Mono Mode

Both channels are combined in anti-phase to produce a single high-power mono output. Connect the speaker between CH A (+) and CH B (–) binding posts. Only one input signal is required — connect to CH A XLR.

Minimum load: 8Ω. Do NOT use loads below 8Ω in bridge mode — risk of severe damage.

Mode	Min. Load	Max Power (TTA1300 example)	Input Used
Stereo	4Ω / ch	1950W × 2 @ 4Ω	CH A + CH B
Bridge Mono	8Ω	3900W @ 8Ω	CH A only
Parallel (4-ch)	4Ω / ch	930W × 4 @ 4Ω	CH A → all ch

11. Gain & Sensitivity Setup

Correct gain structure is critical for maximum headroom and minimum noise floor. The TTA Series input sensitivity is switchable between 0.775V (–2.2 dBu) and 1.44V (+3.2 dBu), with a fixed default gain of 32 dB. Follow this procedure for optimal results:

1. Set all amplifier gain controls to minimum (fully counter-clockwise, position 0).
2. Set your mixer/source to its normal operating level (typically –10 dBu to +4 dBu peaks).
3. Set the Input Sensitivity switch to match your source: 0.775V for consumer/semi-pro equipment, 1.44V for professional +4 dBu gear.
4. Play programme material at the loudest level expected during the performance or event.
5. Slowly raise the amplifier gain until the Signal LED illuminates consistently and the Clip LED flickers only occasionally at peaks.
6. Reduce gain slightly until clipping is rare. This is your optimal gain setting for that installation.
7. Document the gain knob position for future reference and consistent replication at repeated events.

NOTE: Always set gain at the amplifier to match the source level. Never rely solely on high amplifier gain with low source level — this needlessly raises the noise floor. Good gain structure: source near nominal → amplifier provides the remaining gain needed.

12. Protection Systems

The TTA Series includes multi-stage automatic protection to safeguard both the amplifier and connected loudspeakers under all operating conditions:

Protection	Trigger Condition	Indicator	Recovery
Soft-Start	Every power-on	Power LED ramp	Auto — 2–3 sec
Clip Limiting	Output near clipping	CLIP LED flashes	Auto — continuous
DC Offset	DC on output	PROTECT LED on	Auto after fault clears
Short Circuit	Shorted output cable	PROTECT LED on	Fix cable; power cycle
Over-Current	Load below min impedance	PROTECT LED on	Check load; power cycle
Thermal	Heatsink over-temp	PROTECT LED on	Auto after cooling

Soft-Start / Inrush Control

On power-up, the unit delays full output for approximately 2–3 seconds, preventing transient pops that could damage tweeters. The Power LED ramps to full brightness as the output enables.

DC Offset Protection

Continuously monitors the output terminals for any DC voltage (indicating an internal fault). Upon detection, output is immediately muted and the Protect LED illuminates.

Short-Circuit Protection

Detects a shorted speaker cable or connector and immediately mutes the affected channel without damage to the output stage. Inspect all wiring before restarting.

Over-Current Protection

If output current exceeds safe limits — typically caused by driving loads below minimum rated impedance — the current limiter activates to maintain the output stage within safe boundaries.

Thermal Protection (Over-Temperature)

The variable-speed fan automatically increases speed as heatsink temperature rises. If temperature still exceeds the safe threshold, the unit progressively reduces gain and mutes to prevent damage. Allow the unit to cool with mains power ON so the fan continues running.

Clip Limiting

An analogue clip limiter automatically reduces the input level when the output approaches full-scale clipping, providing transparent dynamic gain control and preventing hard clipping distortion that could damage speaker components.

13. Troubleshooting

No output / No power LED

→ Check AC mains connection and verify the power switch is ON.

- Verify fuse in IEC inlet (replace with same type and rating only).
- Check Power LED — if not lit, the unit is not receiving mains power.

Protect LED on / No audio output

- Allow unit to cool for 10 minutes with power ON (fan continues running).
- Check speaker cables for short circuits or bare wire strands across terminals.
- Ensure speaker impedance meets minimum requirements: 4Ω stereo, 8Ω bridge.
- If Protect LED remains on after restart, disconnect all inputs and speaker loads and retry.
- Contact authorised service if the fault persists.

Hum or noise in output

- Check all XLR cable shield connections (Pin 1).
- Try lifting ground on source equipment using a DI box or ground-lift adapter.
- Connect chassis ground screw to rack frame.
- Ensure signal and mains power cables are routed separately.

Clip LED always on

- Reduce source level or amplifier gain.
- Check that the Input Sensitivity switch is set appropriately for the source.
- Ensure the input signal does not exceed +20 dBu.

Distorted sound

- Reduce gain to eliminate clipping.
- Check speaker cables for intermittent shorts.
- Verify speaker impedance — operating below minimum rated impedance causes distortion.

Unit runs very hot / Fan very loud

- Verify adequate rack ventilation (min. 1U above and below the amplifier).
- Check that the rear exhaust is not obstructed.
- Consider reducing output level for thermally constrained rack environments.

14. Maintenance & Care

- Clean the exterior with a dry or slightly damp lint-free cloth. Never use solvents, alcohol, or abrasive cleaners.
- Periodically — every 6 months in dusty environments — use compressed air to clear dust from ventilation slots with power disconnected.
- Inspect power and signal cable connectors for corrosion or physical damage before each use.
- Tighten rack mounting screws if vibration is observed during operation.
- Do not attempt to lubricate or service the internal fan — this must be done by authorised service personnel.
- Store in the original packaging or a protective flight case when transporting.
- If the unit has not been used for more than 6 months, power it on briefly (without load) before use to re-form electrolytic capacitors.
- Check binding post connections periodically for tightness and oxidation; clean with contact cleaner if necessary.

15. Warranty & Support

Louis Martin Audio warrants this product against defects in materials and workmanship for a period of **2 years** from the date of original purchase by the end-user from an authorised dealer. This warranty is valid for the original purchaser only and is non-transferable.

Warranty Exclusions:

- Damage resulting from misuse, accidents, or modifications not authorised by Louis Martin Audio.
- Damage caused by operation outside specified voltage or impedance ranges.
- Cosmetic damage (scratches, dents) that does not affect functionality.
- Consumable parts such as fuses and contact cleaners.
- Equipment used in environments not compliant with the safety instructions in Section 1.
- Units with removed or altered serial number labels.

Contact & Support:

Website	www.louismartinaudio.it
Email	contact@louismartinaudio.it
Service	Authorised service centres listed on the website

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